

SAI KARTHIK MADDIRALA

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SUMMARY

AI/ML Engineer specializing in, Generative AI, and LLM agents. Build AR virtual try-on and retrieval-augmented (RAG) systems with PyTorch, LangChain, and Claude/OpenAI APIs, owning the lifecycle from training through real-time inference, deployment, and monitoring.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, TypeScript, Node.js, React, Bash

Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, Deep Learning, Model Evaluation, A/B Testing

Generative AI and LLMs: Claude API, OpenAI API, LangChain, LangGraph, Hugging Face, Ollama, RAG, Hybrid Retrieval, Reranking, Prompt Engineering, AI Agents, RAGAS

Backend and APIs: FastAPI, Flask, REST APIs, Streaming APIs (NDJSON/SSE), Microservices

Databases and Vector Stores: PostgreSQL, MySQL, MongoDB, SQLite, ChromaDB, FAISS, Pinecone

MLOps and Cloud: Docker, Kubernetes, MLflow, CI/CD, GitHub Actions, AWS (SageMaker, S3, Lambda), Databricks

EXPERIENCE

ML Engineer

Mar 2026 – Present

CDF (Remote)

Kansas, USA

- Own the end-to-end ML lifecycle for an augmented-reality virtual try-on feature – dataset curation and labeling, training, evaluation, deployment, and monitoring – taking models from prototype to production on a live camera feed.
- Built PyTorch training and fine-tuning pipelines with augmentation, loss tuning, and hyperparameter search, lifting model accuracy from 79% to 88% (0.89 mIoU) on a 5K-image validation set.
- Cut serving latency 45% (110ms to 60ms) and model size 65% through INT8 quantization, pruning, and ONNX/TensorRT export, sustaining 25–30 FPS real-time inference on mid-range mobile devices.
- Built MLOps tooling – automated labeling, training, and evaluation with MLflow experiment tracking – cutting the retrain-and-evaluate cycle from 2 days to 6 hours.
- Deployed containerized models on Kubernetes with data-drift monitoring, and validated releases through A/B testing, driving a 14% lift in product-page engagement in a pilot rollout.

Machine Learning Intern

Jun 2021 – Aug 2021

Hex N Bit

India

- Delivered a loan approval prediction model reaching 0.86 ROC-AUC, benchmarking Logistic Regression, Random Forest, and XGBoost across 5-fold cross-validation.
- Lifted default-class recall from 68% to 81% by engineering features from income, credit history, and employment attributes and countering class imbalance with SMOTE resampling.
- Tuned decision thresholds to reduce projected default risk 12% at a comparable approval rate, and used feature-importance analysis to explain predictions to reviewers.

PROJECTS

Voyager: Agentic AI Road-Trip Planner | *Claude API, AI Agents, Tool Use, Flask, Python* | [GitHub](#)

- Engineered a two-agent system on the Claude API where a planning agent runs its own tool loop, deciding when to call tools for information and drive times, then picking stops that fit the user's days – no hardcoded call points.
- Built a copilot agent that reads the live plan each turn and decides whether to act or advise, executing edits like “add Vail, drop Denver” and re-optimizing the route.
- Eliminated hallucinated recommendations by grounding every rating, distance, and forecast in live APIs (Google Places, OSRM, Open-Meteo) with deterministic routing pinning the drivable order.

Browsing History RAG | *Hybrid Retrieval, RRF, Cross-Encoder Reranking, ChromaDB, Ollama, FastAPI, LLMops* | [GitHub](#)

- Built a fully local RAG system answering plain-English questions over Chrome browsing history, with an LLM query router deciding per turn whether to retrieve or answer from memory (adaptive RAG), keeping 100% of data on-device.
- Boosted retrieval quality with a hybrid pipeline: dense ChromaDB vectors and BM25, fused by Reciprocal Rank Fusion and reranked by a cross-encoder to a top-15 context, plus time-aware path filtering windows like “last week” before ranking.
- Streamed answers token-by-token from a local Ollama LLM via a FastAPI NDJSON endpoint into a React chat UI, and drove tuning with an offline eval harness (recall/precision/MRR, RAGAS, LLM-as-judge faithfulness) in MLflow.

EDUCATION

The University of Kansas

M.S. in Computer Science

Aug 2023 – Dec 2025

Lawrence, KS